

Network Management

Khaula Molapo

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Network Management Plan (Medium-Sized Business)

Environment Overview:

- 3 office locations
- 500 employees
- Hybrid cloud environment
- Remote workforce (VPN access)

1. Monitoring Strategy & Tools

Strategy:

- Centralized monitoring system
- Real-time alerts and reporting
- Monitor devices, servers, and applications

Tools:

- SolarWinds / PRTG
- AWS CloudWatch / Azure Monitor
- ELK Stack

Metrics:

- Latency, uptime, CPU usage, bandwidth

2. Security Management Procedures

- Role-Based Access Control (RBAC)
- Multi-Factor Authentication (MFA)
- Firewalls and VPN access
- Network segmentation (VLANs)
- IDS/IPS systems
- Regular patch updates

3. Performance Optimization Plan

- Load balancing
- Bandwidth management (QoS)
- Caching mechanisms
- Network upgrades when needed
- Cloud auto-scaling

4. Disaster Recovery Procedures

- Daily backups (cloud + on-site)
- Data replication across sites
- Failover to secondary systems
- RTO: 2 hours
- RPO: 15 minutes
- Regular testing

5. Budget & Resource Allocation

- Monitoring tools: 20%
- Security: 30%

- Hardware: 25%
- Cloud: 15%
- Training: 10%

Resources:

- Network Admins (2-3)
- Security Specialist (1)
- IT Support (3-5)

Troubleshooting Scenario Analysis

Scenario 1: Intermittent Application Slowness

Root Cause:

- Log analysis, latency checks, traffic spikes

Steps:

1. Check bandwidth
2. Test latency
3. Review server performance
4. Analyze logs
5. Isolate issue

Solutions:

- QoS implementation
- Bandwidth upgrade
- App optimization

Prevention:

- Monitoring
- Traffic control
- Capacity planning

Scenario 2: Wireless Issues

Root Cause:

- Interference, overload, channel issues

Steps:

1. Scan interference
2. Check devices
3. Adjust AP placement
4. Change channels
5. Test signal

Solutions:

- Add access points
- Upgrade to Wi-Fi 6
- Load balancing

Prevention:

- Site surveys
- Regular audits
- Proper planning

Scenario 3: Security Breach

Root Cause:

- Log analysis, identify entry point

Steps:

1. Isolate systems
2. Disable accounts
3. Scan malware
4. Review logs
5. Patch issues

Solutions:

- Stronger firewall
- Enforce MFA
- Improve IDS/IPS

Prevention:

- User training
- Vulnerability scans
- Penetration testing
- Continuous monitoring